

The General Theory of Systems

BY

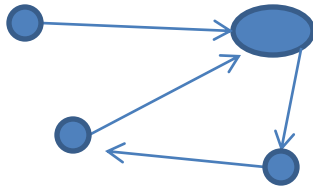
ROBERT JOHANNSON

The Two Systems

There are two different definitions of a system:

Elemental Systems.

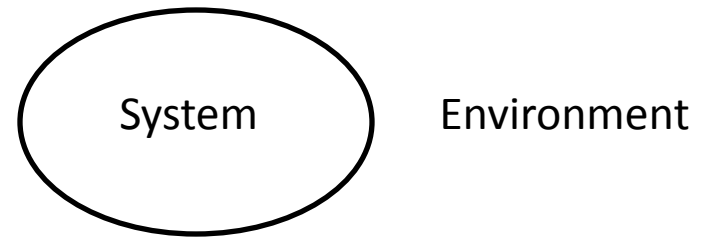
A system is the pattern of relationships between a number of elements.



The elements are considered real and the system is merely an aspect of the elements. Thus physics, chemistry and mathematics.

Holistic Systems

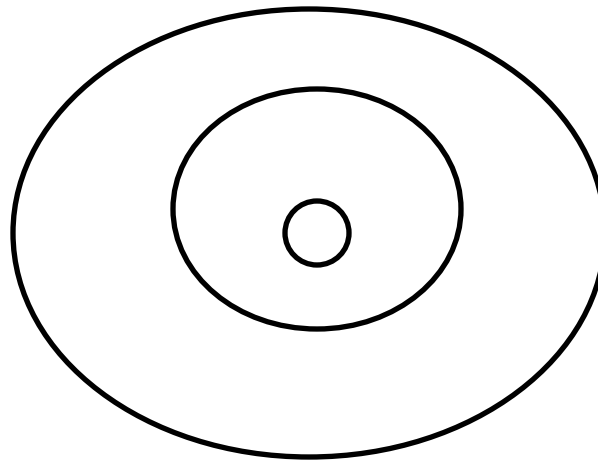
A system is a boundary that separates the universe into the system and its environment.



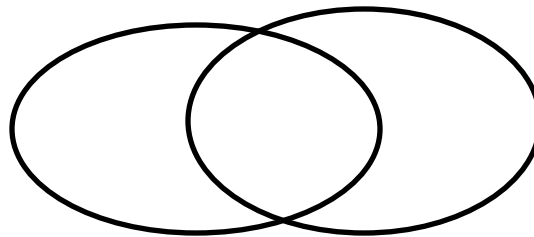
Wholes have parts that take their identity from the whole. E.g. The lung is a part of the body. Outside the body the lung is just dead meat.

Holarchy

Holistic systems are hierarchical. As a whole is made up of parts, so each of the parts is a whole to its own parts. Arthur Koestler coined the term “holon” to describe this dual quality of wholes.



Since wholes are logical categories they also intersect to form complex wholes.



Two Different Aspects of Reality

The reason there are two different definitions of systems is because there are two different aspects of reality.

BODY

Matter, atoms,
molecules,
energy, forces

MIND

Ideas, decisions,
information,
communication,
conversations,
most of human life

BASIC UNITS

BODY: Atoms

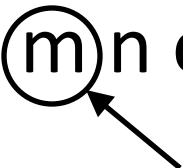
MIND: Information

The Technical Definition of Information

The engineers at Bell Labs developed a simple definition of information:

Information is a message selected from a set of possible messages.

Possible messages:

a b c d e f g h i j k l  n o p q r s t u v w x y z

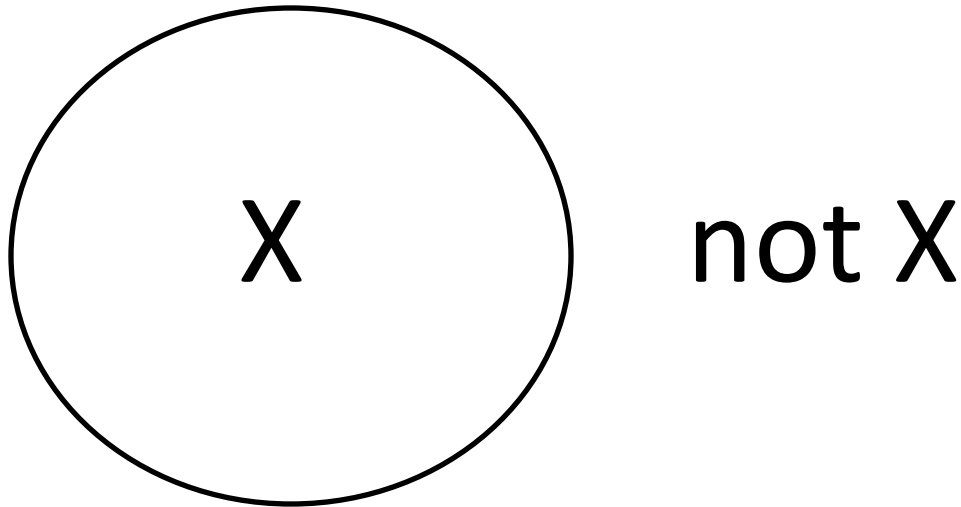
Information: the choice

The Basic Unit of Choice

The basic unit of choice is the binary choice: either/or.

This choice is the basis of Boolean logic.

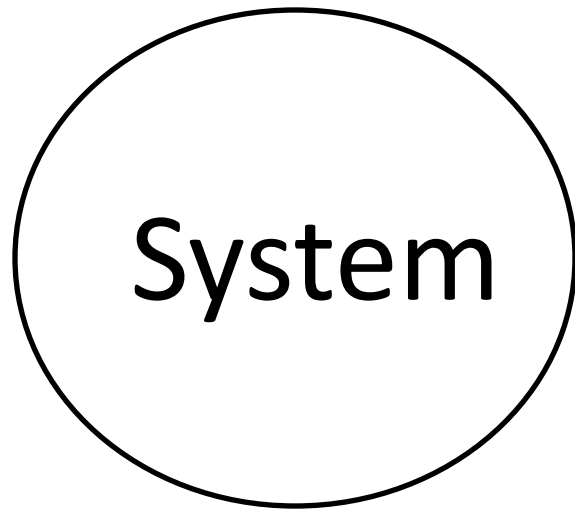
It is symbolized by the Venn diagram where the choice is represented by a line separating the x from the not x . It can also be used to diagram the syllogisms in traditional Aristotelian categorical logic.



Basic Binary Choice

Information and Systems

Holistic systems are a function of information.



Environment

The Binary Code

Any complex choice can be coded as a series of binary choices, usually symbolized by 0 and 1.

You can calculate the number of binary digits or “bits” needed to code a set of possibilities by taking the logarithm to the base 2 of the number of possibilities. Thus with 26 letters and a space we have $\text{Log}_2 27 = 5$ bits. Thus using the binary either/or as our basic unit of information we can count the amount of information by counting the number of bits it takes to code it.

Thus

A = 00000

B = 00001

C = 00010

D = 00011

E = 00100

F = 00101

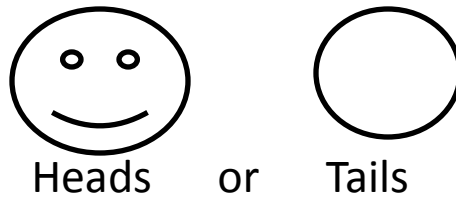
Etc.

This is the basis of modern computing where the computer is a chip with transistors that can be used as a gate either open or closed. Thus the binary code becomes a series of transistor positions. Computers are programmed with Boolean logic.

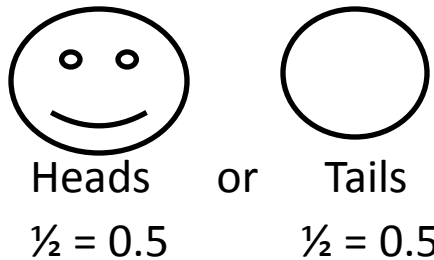
Possibilities and Probabilities

Since both information and statistical probabilities are calculated from a set of possibilities there is often confusion about the difference. Some people even think that information is a probability function. The difference is that information is measured in bits and probabilities are measured in fractions.

A coin has two possibilities.

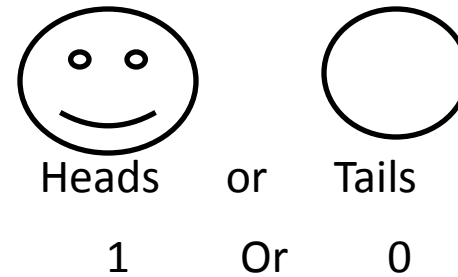


Probabilities



Probabilities are a fraction.

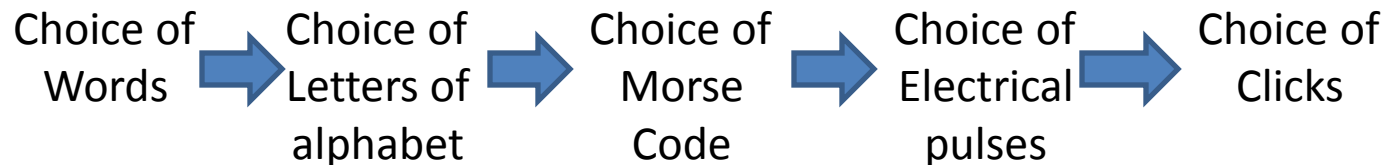
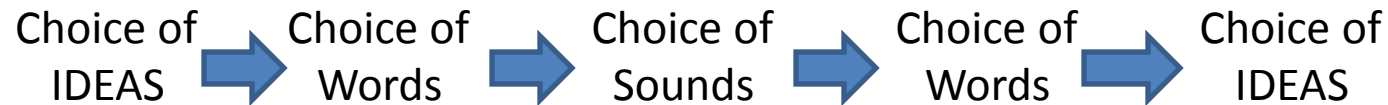
Information



Information is measured in bits

Communication as Translation

Since information is choice within a set, communication consists of translating choices in one set of possibilities into choices in another set, in other words coding.



Although information consists of choices in a set and has no meaning as such. The meaning is in the reference. Thus the map is not the territory, but the territory is the meaning of the map. Thus the clicks mean electrical pulses which mean Morse code which mean letters which mean words which mean ideas.

The study of information and control is called “cybernetics.”

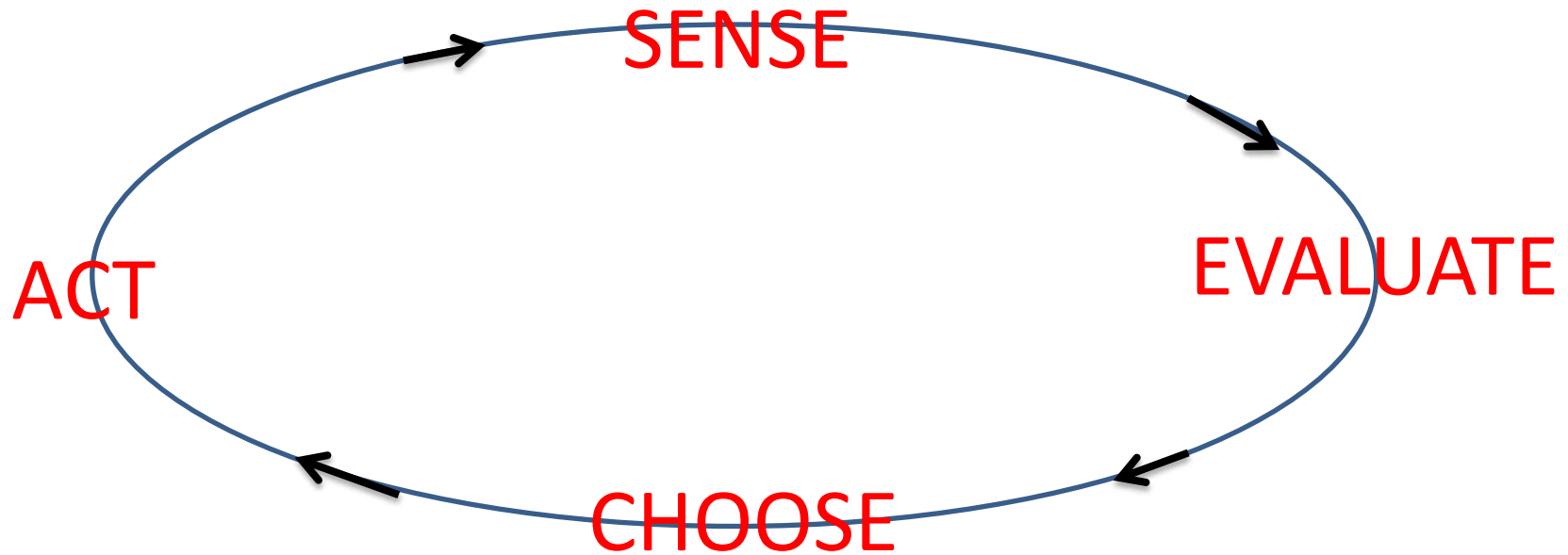
The term “cybernetics” comes from the Greek word “KYBERNĀΩ” which means to steer, to guide or to govern.

In plain English it is the theory of control.

We are all familiar with the basic control process.

1. (ACT) You do something.
2. (SENSE) You see how it turned out.
3. (EVALUATE) You compare the results to what you wanted.
4. (CHOOSE) You choose another action to bring it closer to your goal.
5. (ACT) You do something else.

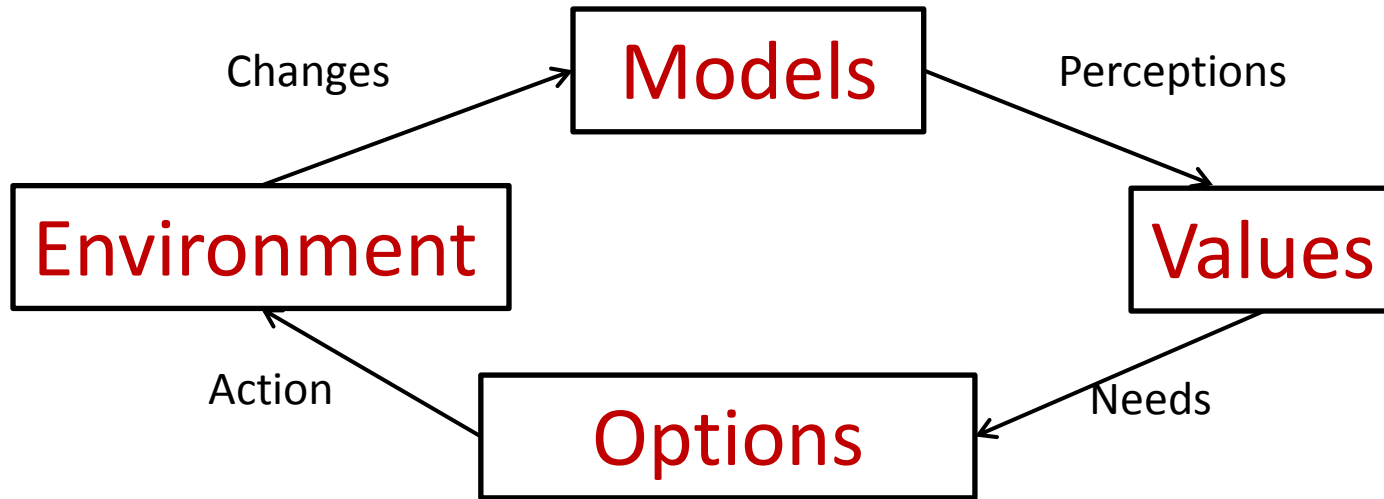
The Cybernetic Control Loop



These four processes form a loop. Any control process consists of going around this basic control loop.

We can use the box as a symbol of the translation process, and the arrow as a symbol of the information.





Since information is a decision, it can also act as a command. The translation processes are commands about how to process information, in other words programs. The translation processes constitute the program that controls the control process.

“What is real?”

Western Materialism teaches that only matter and energy are objectively real. But most of human life consists of ideas, decisions, information, communication, and conversations. If human life is defined as not real, then this is a problem, the mind/body problem.

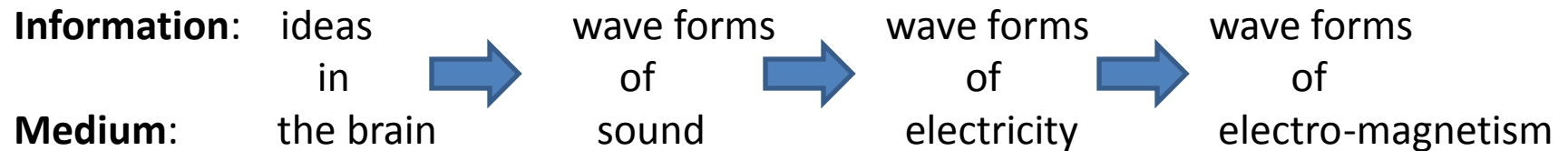
Fortunately we have engineers and they came up with two solutions.

1. The Telecommunications Solution

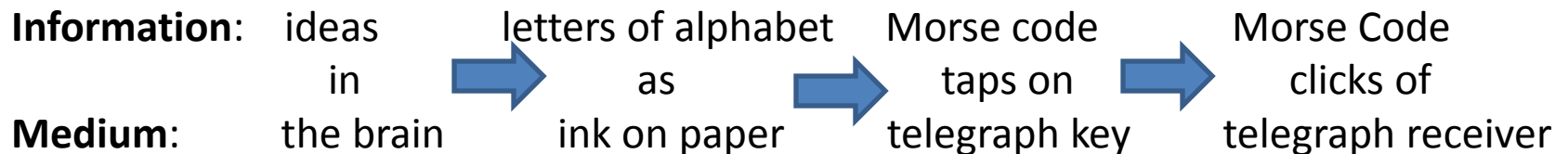
MEDIA

**Matter and energy are the media
of information and communication.**

RADIO



Telegraph



2. The Cybernetics Solution Control

Information and communication control
matter and energy.

The brain processes
Information and
controls



The muscles
which process
energy

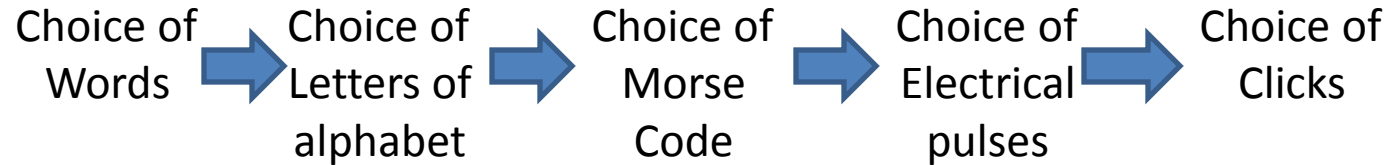
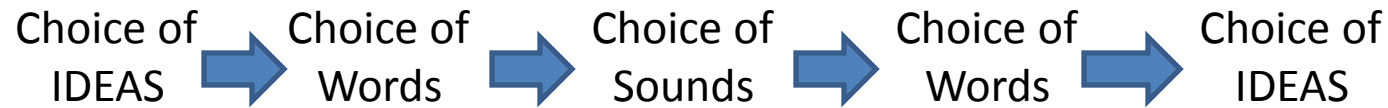
The thermostat processes
Information and controls



A heater which
processes energy

Two Different Processes

Communication is Translation



Matter/Energy is a Constant

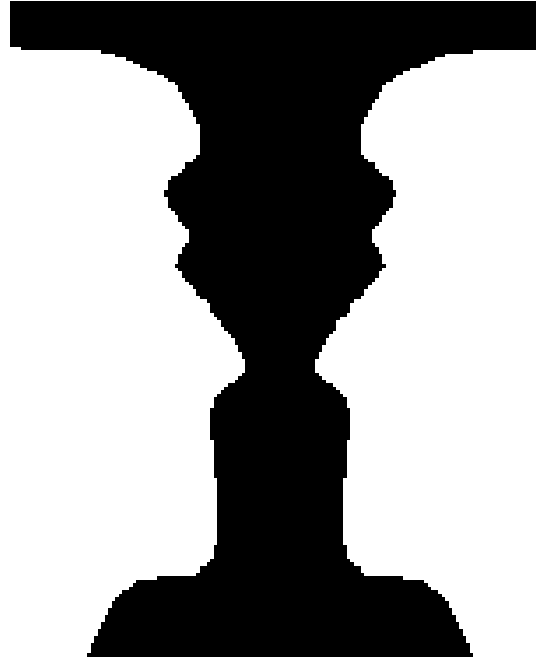
$$K_f + U_f = K_i + U_i$$

$$E = mc^2$$

$$P = E/t$$

The two processes are related as a figure/ground gestalt.

Figure and Ground



We can choose which is the figure and which is the ground.

If you choose the black as the background, and the white as the figure, then you will see two faces. If you choose the white as the background and the black as the figure then you will see a vase.

Mind and Body

Figure/Ground

**Matter and energy are the
media**

**[Background for the figure]
of information and
communication.**

**Information and communication
control**

**[Background for the figure]
matter and energy.**

Examples

**Paper and screen are the
media**

**[Background for the figure]
Of the printed word**

**Mathematics
controls**

**[Background for the figure]
Molecules in motion**

Holistic Understanding

Holistic understanding comprehends both the information process and the energy process.

DNA

DNA is an acid
(a molecular energy process)
which acts as a
medium
for genetic information which
controls
the material development of the organism.

Predicting the Future

Matter and energy are constants therefore matter/ energy processes are predictable.

Information is a decision therefore information processes are only partly predictable.

Programmed decisions are predictable, but the program can be changed.

The future is not perfectly predictable because it depends on decisions that haven't been made yet.